

HORNSBY GIRLS HIGH SCHOOL



Mathematics Advanced

Year 11 Assessment 3 Yearly Examination Term 3 2019

STUDENT NUMBER: _____

General Instructions

- Reading Time – 10 minutes
- Working Time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators and drawing templates may be used
- In Questions 11 – 15, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for untidy and poorly arranged work
- Do not use correction fluid or tape
- Do not remove this paper from the examination

Total marks – 77

Section I Pages 2 – 5

10 marks

Attempt Questions 1 – 10

Answer on the Objective Response Answer Sheet provided

Section II Pages 6 – 11

67 marks

Attempt Questions 11 – 16

Question	1-10	11	12	13	14	15	16	Total
Total	/10	/11	/11	/12	/11	/11	/11	/77

This assessment task constitutes 40% of the Preliminary Higher School Certificate Course School Assessment

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the Objective Response answer sheet for Questions 1 – 10

1 Which one, of the following sets of points does NOT represent a function?

(A) $\{(-6, 4), (-5, 1), (-1, 6), (0, 5), (6, 3)\}$

(B) $\{(-3, 0), (4, 2), (-1, 1), (0, -3), (4, 3)\}$

(C) $\{(-3, 0), (-5, 2), (-1, 0), (1, -3), (4, 3)\}$

(D) $\{(0, 0), (\pm 1, 0), (\pm 2, \sqrt{3}), (\pm 3, \sqrt{8})\}$

2 Which expression is NOT equivalent to $a^{-1\frac{1}{2}}$?

(A) $\frac{1}{a^2\sqrt{a}}$

(B) $a^{-\frac{3}{2}}$

(C) $\frac{1}{a\sqrt{a}}$

(D) $\frac{\sqrt{a}}{a^2}$

3 Which of the following is an even function?

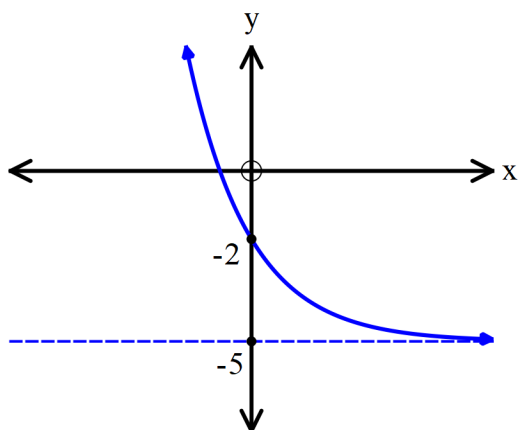
(A) $y = x^3 + x$

(B) $y = |x + 1|$

(C) $y^2 = 9 - x^2$

(D) $y = |x| - 1$

- 4 Which of the following functions represent the one shown in the graph?



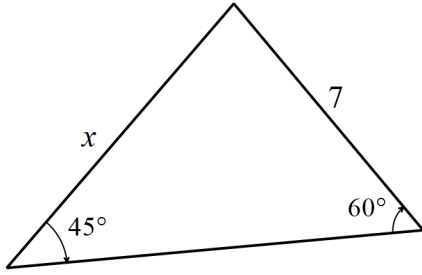
- (A) $y = 2^{-x} - 2$
- (B) $y = 2^{-x} - 5$
- (C) $y = 3 \times 2^{-x} - 5$
- (D) $y = 3 \times 2^{-x} - 2$
- 5 What are the solutions of $|x^2 - 3| = 6$

- (A) 3 and -3
- (B) 3 and $\sqrt{3}$
- (C) ± 3 and $\pm\sqrt{3}$
- (D) $\sqrt{3}$ and $-\sqrt{3}$

- 6 A horizontal line is tangent to $y = 2x^2 + 2$ at point P . The coordinates of P are:

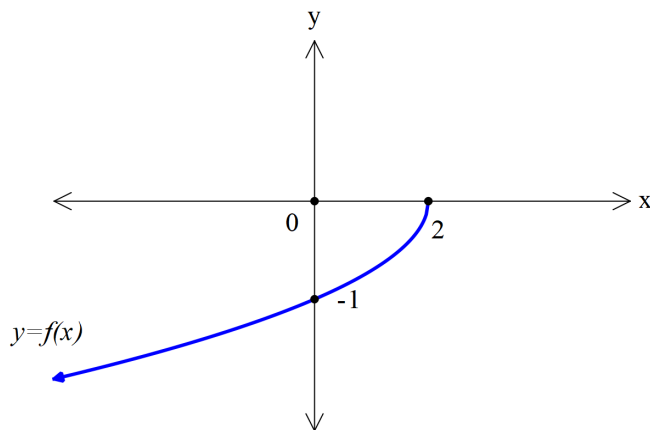
- (A) (0, 0)
- (B) (0, 2)
- (C) (0, -2)
- (D) (2, 0)

- 7 Find x , expressing your answer as an exact value:



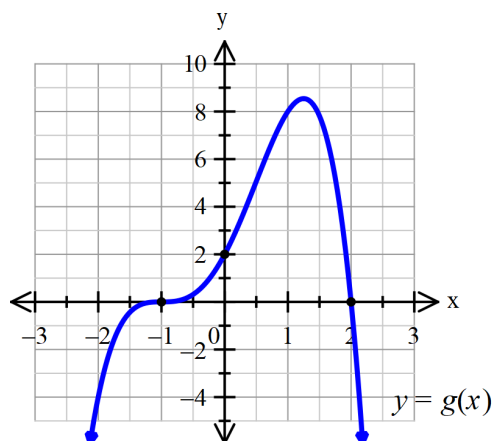
- (A) $\frac{7\sqrt{6}}{2}$
- (B) $\frac{\sqrt{6}}{14}$
- (C) $\frac{7\sqrt{2}}{2}$
- (D) $\frac{7\sqrt{6}}{3}$

- 8 What is the domain $D(f)$ and range $R(f)$ of the following function:



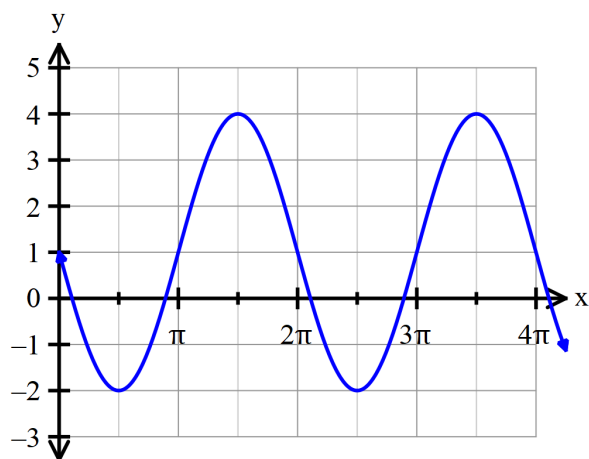
- (A) $D(f) = (2, \infty), R(f) = (-\infty, 0)$
- (B) $D(f) = (-\infty, 0], R(f) = (-\infty, 2)$
- (C) $D(f) = (-\infty, 2], R(f) = (-\infty, 0]$
- (D) $D(f) = [2, -\infty), R(f) = [0, -\infty)$

- 9 Which of the following functions represents the graph shown below?



- (A) $g(x) = (x-1)^2(x-2)$
- (B) $g(x) = (x+1)^3(x-2)$
- (C) $g(x) = -(x+1)^2(x-2)^2$
- (D) $g(x) = -(x+2)^3(x-1)$

- 10 Which of the following functions represents the graph shown below?



- (A) $y = 3 \sin x + 1$
- (B) $y = 3 \cos x - 1$
- (C) $y = 1 - 3 \sin x$
- (D) $y = 1 - 3 \cos x$

End of Section I

Section II

67 marks

Attempt Questions 11-16

Question 11 (11 marks)

- (a) Simplify $\frac{x^{-1} - x^{-2}}{x - x^2}$, giving your answer in fraction form with positive indices. 2

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- (b) Solve $\log_2(x^3) + 5 = 11$. 2

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- (c) Given $\cos \theta = -\frac{2}{3}$, and $\tan \theta > 0$, find the EXACT value of $\sin \theta$. 2

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- (d) Solve $6^x + \frac{36}{6^x} = 37$. Leave your answer in exact form. 2

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- (e) A circle has diameter AB, where $A = (2, 5)$ and $B = (-4, 1)$.

- (i) Find the radius of the circle in simplest surd form. 1

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- (ii) Find the centre of the circle. 1

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- (iii) Hence, write the equation of the circle. 1

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Question 12 (11 marks)

- (a) Use first principles , $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$, to find the gradient function of

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$$f(x) = \frac{1}{x} .$$

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- (b) Prove that $\sin^2 x \tan^2 x + 2 \sin^2 x + \cos^2 x = \sec^2 x$.

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- (c) Solve $\operatorname{cosec} (\theta - 30^\circ) = 3$ for $0^\circ \leq \theta \leq 180^\circ$. Answer correct to the nearest degree.

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(d) Given $f(x) = \begin{cases} 1-x^2, & \text{for } x \leq 1 \\ kx+1, & \text{for } x > 1 \end{cases}$

- (i) Find the value of k for which $f(x)$ is continuous at $x=1$. 1

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- (ii) If $k = -1$, find the value of $f(-3) + f(2)$. 1

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- (iii) If $k = -1$, find the range of $f(x)$. 1

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- (e) If the minimum value of parabola $y = x^2 - 3x + k$ is 5, find the value of k . 2

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Question 13 (12 marks)

- (a) Differentiate the following functions, leaving your answers in simplest factorised form where possible:

(i) $y = \frac{2}{\sqrt{x}}$ 2

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(ii) $y = (e^{2x} - 1)^3$ 2

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(iii) $f(x) = x^3 \sqrt{x^2 - 1}$ 2

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(b) Consider the function $f(x) = \frac{1}{3}x^3 - x^2 - 15x + 3$.

(i) Find $f'(x)$.

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(ii) Find the values of x such that $f'(x) = 0$.

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(iii) Find the set of values of x for which $f(x)$ is decreasing.

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(iv) Find the gradient of the tangent and normal of $f(x)$ at $x = 0$.

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Question 14 (11 marks)

- (a) Consider the equation $x + \frac{2}{x} = p$, find the values of p for which the equation has real roots. **2**

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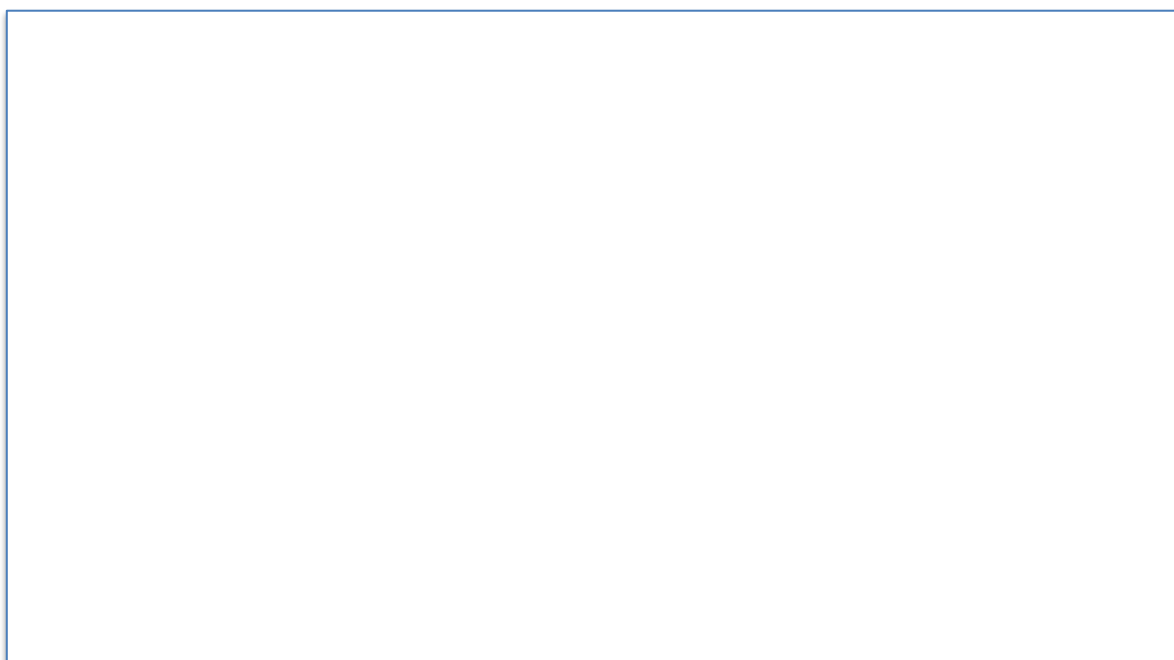
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- (b) For $f(x) = |x + 2| - 3$,

- (i) Sketch the function, showing any intercepts. **3**



- (ii) Hence, or otherwise, find the values for $f(x) \geq 1$. **1**

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- (c) A bacteria replicated itself according to the equation $N(t) = 140 \times 2^{3.5t}$, where $N(t)$ is the number of bacteria present in a test tube t hours after the experiment began.

- (i) What is the initial number of bacteria added to the test tube? 1

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- (ii) How many bacteria are there after 10 hours? 2
 Answer in scientific notation correct to 3 significant figures.

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- (iii) When the number of bacteria reaches 3×10^{25} in the test tube, it will stop replicating. How long before this number is reached? 2
 Give your answer correct to the nearest minute.

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Question 15 (11 marks)

(a) Given that $f(x) = \frac{2x+1}{3x-4}$, $\left(x \neq \frac{4}{3}\right)$, find $f'(-2)$.

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(b) Solve $\log_2(x+4) + \log_2(x-1) - \log_2(x+2) = 2$.

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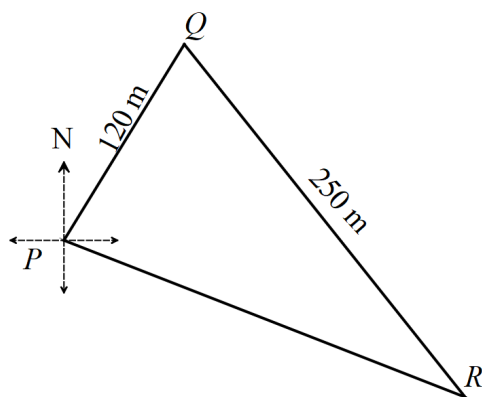
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- (c) A farmer walked 120 metres from P to Q on a bearing of 040°T .
He then walked 250 metres from Q to R on a bearing of 140°T .



- (i) Find the value of $\angle PQR$. 1

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- (ii) Find the length of PR , correct to the nearest metre. 2

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- (iii) Find the bearing of R from P , correct to the nearest degree. 2

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- (iv) Find the area of $\triangle PQR$. Answer correct to the nearest m^2 . 1

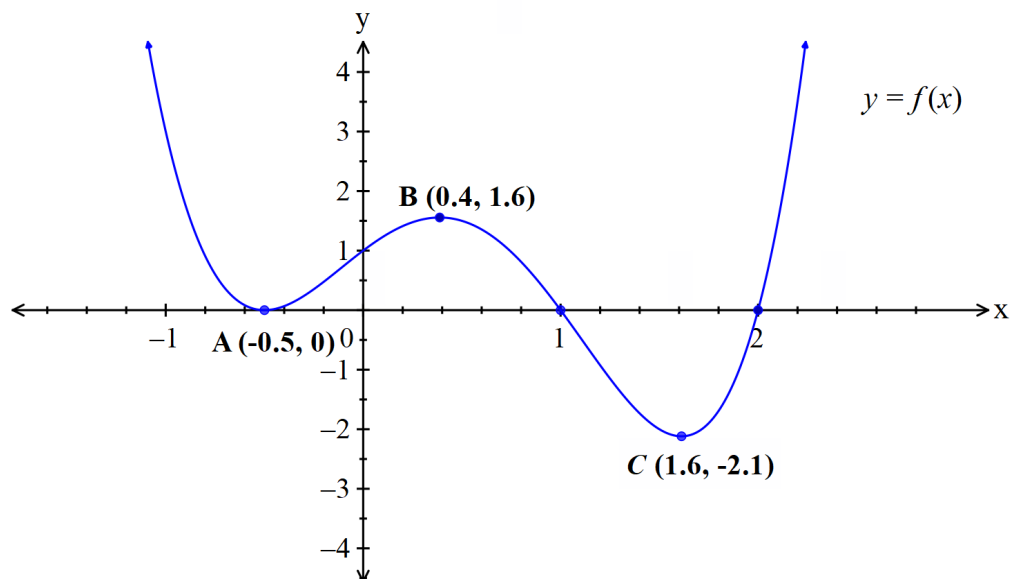
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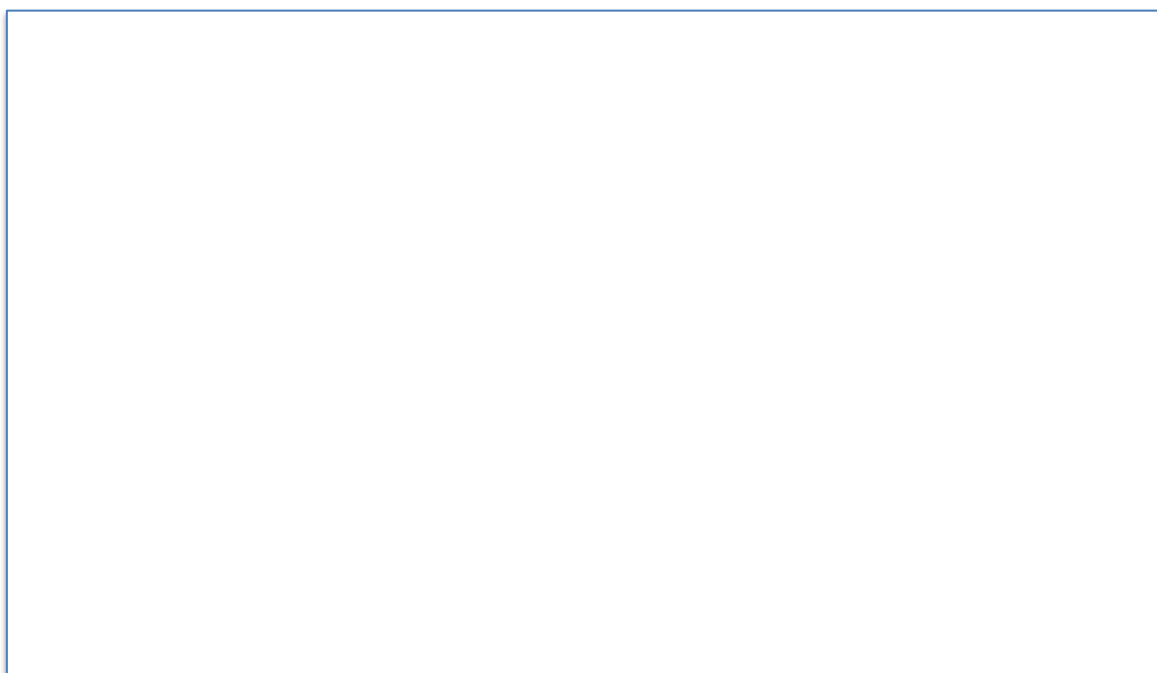
Question 16 (11 marks)

- (a) The diagram below shows the graph of a function $y = f(x)$. Point A, B and C are stationary points at which the derivatives are zero.



- (i) Sketch the graph of $f'(x)$. Clearly label all essential information.

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- (ii) State the values of x for which $f(x)$ is increasing.

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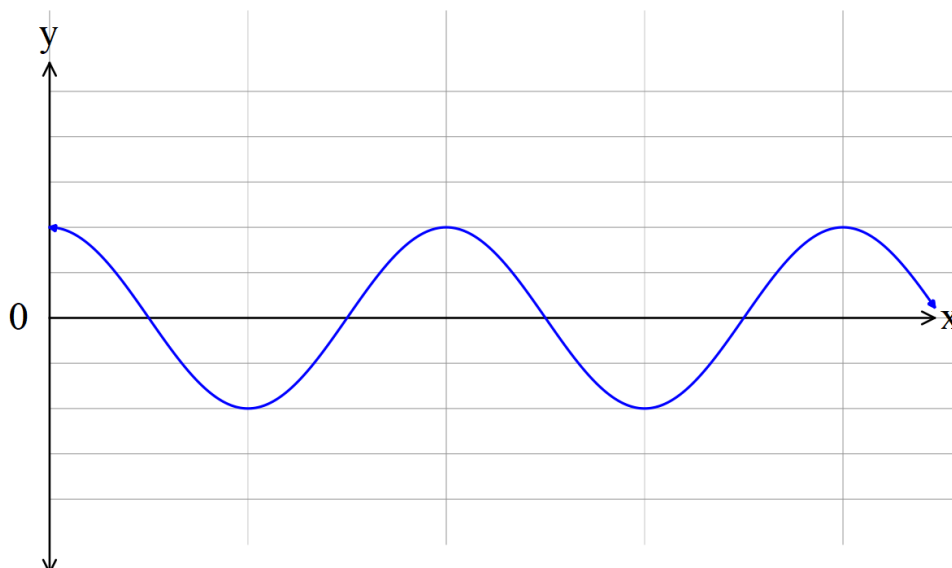
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- (b) The graph of $y = 2 \cos(2x)$ is shown below, where x is in radians.



- (i) Find the period and amplitude. 2

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- (ii) On the above graph, label all the x - and y - intercepts on both axes. 2

- (iii) Draw $y = 2 \sec(2x)$ on the above graph, clearly showing any asymptotes. 1

- (iv) Find the exact solutions for $2 \sec(2x) = 4$ for $0 \leq x \leq 2\pi$. 2

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End of Examination

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HORNSBY GIRLS HIGH SCHOOL



Mathematics Advanced

Year 11 Assessment 3 Yearly Examination Term 3 2019

STUDENT NUMBER: _____ **SOLUTION** _____

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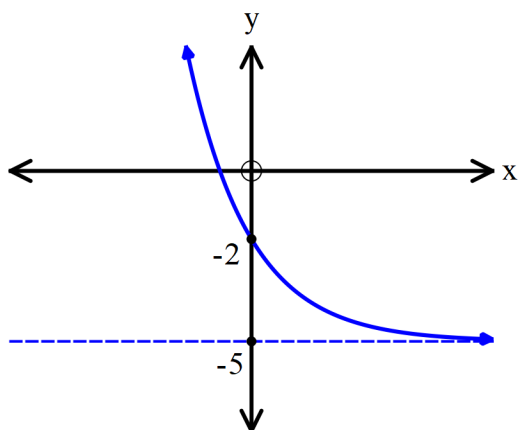
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- (B) $y = 2^{-x} - 5$
- ☒ (C) $y = 3 \times 2^{-x} - 5$
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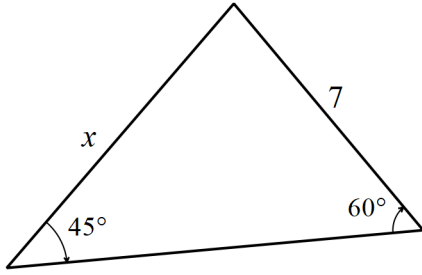
- 5 What are the solutions of $|x^2 - 3| = 6$

- ☒ (A) 3 and -3
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- 6 A horizontal line is tangent to $y = 2x^2 + 2$ at point P . The coordinates of P are:

- (A) (0, 0)
- ☒ (B) (0, 2)
- (C) (0, -2)
- (D) (2, 0)

- 7 Find x , expressing your answer as an exact value:



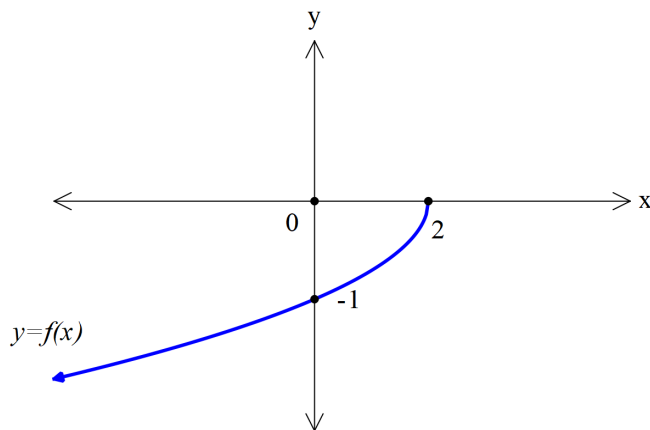
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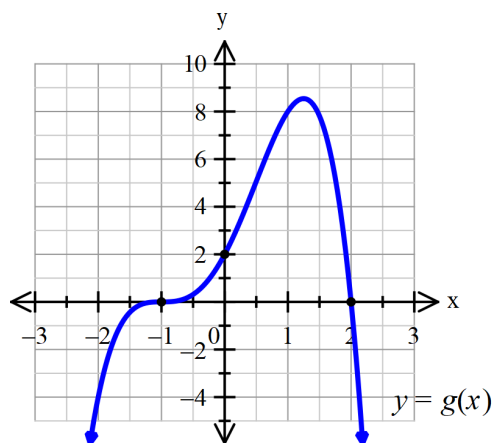
(A) $D(f) = (2, \infty), R(f) = (-\infty, 0)$

(B) $D(f) = (-\infty, 0], R(f) = (-\infty, 2)$

(C) $D(f) = (-\infty, 2], R(f) = (-\infty, 0]$

(D) $D(f) = [2, -\infty), R(f) = [0, -\infty)$

- 9 Which of the following functions represents the graph shown below?



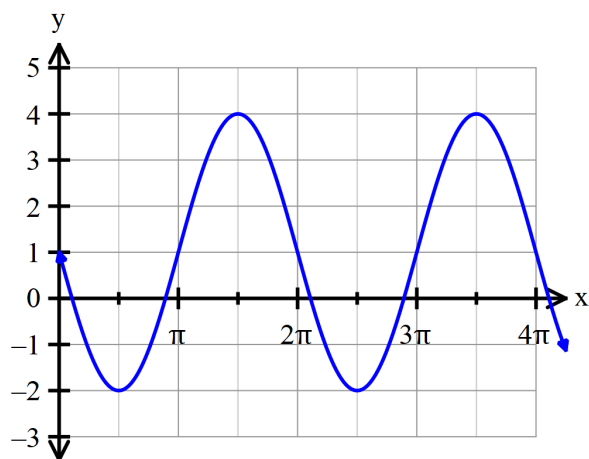
(A) $g(x) = -(x-1)^2(x-2)$

(B) $g(x) = -(x+1)^3(x-2)$

(C) $g(x) = -(x+1)^2(x-2)^2$

(D) $g(x) = -(x+2)^3(x-1)$

- 10 Which of the following functions represents the graph shown below?



(A) $y = 3 \sin x + 1$

(B) $y = 3 \cos x - 1$

(C) $y = 1 - 3 \sin x$

(D) $y = 1 - 3 \cos x$

End of Section I

Section II

67 marks

Attempt Questions 11-16

Question 11 (11 marks)

- (a) Simplify $\frac{x^{-1} - x^{-2}}{x - x^2}$, giving your answer in fraction form with positive indices. 2

$$\begin{aligned}\frac{x^{-1} - x^{-2}}{x - x^2} &= \frac{\frac{1}{x} - \frac{1}{x^2}}{x - x^2} \\ &= \frac{\frac{x-1}{x^2} \times x^2}{(x-x^2) \times x^2} \\ &= \frac{x-1}{x^3(1-x)} \\ &= \frac{-1}{x^3}\end{aligned}$$

- (b) Solve $\log_2(x^3) + 5 = 11$. 2

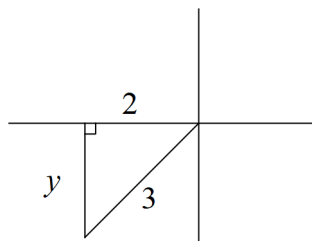
$$\log_2(x^3) = 6$$

$$x^3 = 2^6$$

$$x^3 = 4^3$$

$$x = 4$$

- (c) Given $\cos \theta = -\frac{2}{3}$, and $\tan \theta > 0$, find the EXACT value of $\sin \theta$. 2



$$y^2 = 3^2 - 2^2$$

$$|y| = \pm\sqrt{5}$$

since θ is in Q3

$$\therefore \sin \theta = \frac{-\sqrt{5}}{3}$$

- (d) Solve $6^x + \frac{36}{6^x} = 37$. Leave your answer in exact form. 2

$$(6^x)^2 + 36 - 37 \times 6^x = 0$$

$$(6^x - 36)(6^x - 1) = 0$$

$$(6^x - 36) = 0, (6^x - 1) = 0$$

$$6^x = 36, 6^x = 1$$

$$x = 2, 0$$

- (e) A circle has diameter AB, where $A = (2, 5)$ and $B = (-4, 1)$.

- (i) Find the radius of the circle in simplest surd form. 1

$$\text{diameter} = \sqrt{(-4 - 2)^2 + (1 - 5)^2}$$

$$= \sqrt{36 + 16}$$

$$= \sqrt{52}$$

$$= 2\sqrt{13}$$

$\therefore$ the radius is $\sqrt{13}$ units.

- (ii) Find the centre of the circle. 1

$$M_{AB} = \left(\frac{2 - 4}{2}, \frac{5 + 1}{2} \right)$$

$\therefore$ the midpoint is $(-1, 3)$.

- (iii) Hence, write the equation of the circle. 1

$\therefore$ the equation of the circle is $(x + 1)^2 + (y - 3)^2 = 13$.

Question 12 (11 marks)

- (a) Use first principles, $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$, to find the gradient function of

2

$$f(x) = \frac{1}{x}.$$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \left(\frac{\frac{1}{x+h} - \frac{1}{x}}{h} \right) \\ &= \lim_{h \rightarrow 0} \left(\frac{x - (x+h)}{x(x+h)h} \right) \\ &= \lim_{h \rightarrow 0} \left(\frac{-1}{x(x+h)} \right) \\ &= \frac{-1}{x^2} \end{aligned}$$

- (b) Prove that $\sin^2 x \tan^2 x + 2 \sin^2 x + \cos^2 x = \sec^2 x$.

2

$$\begin{aligned} LHS &= \sin^2 x \frac{\sin^2 x}{\cos^2 x} + 2 \sin^2 x + \cos^2 x \\ &= \frac{\sin^4 x + 2 \sin^2 x \cos^2 x + \cos^4 x}{\cos^2 x} \\ &= \frac{(\sin^2 x + \cos^2 x)^2}{\cos^2 x} \\ &= \frac{1}{\cos^2 x} \\ &= \sec^2 x \\ &= RHS \end{aligned}$$

$$\begin{aligned} \text{Or : } LHS &= \sin^2 x (\sec^2 x - 1) + 2 \sin^2 x + \cos^2 x \\ &= \sin^2 x \sec^2 x - \sin^2 x + 2 \sin^2 x + \cos^2 x \\ &= \frac{\sin^2 x}{\cos^2 x} + \sin^2 x + \cos^2 x \\ &= \tan^2 x + 1 \\ &= \sec^2 x \\ &= RHS \end{aligned}$$

- (c) Solve $\operatorname{cosec}(\theta - 30^\circ) = 3$ for $0^\circ \leq \theta \leq 180^\circ$. Answer correct to the nearest degree.

2

$$\operatorname{cosec}(\theta - 30^\circ) = 3$$

$$\sin(\theta - 30^\circ) = \frac{1}{3}$$

$$-30^\circ \leq \theta - 30^\circ \leq 150^\circ$$

$$\theta - 30^\circ = 19^\circ, 161^\circ$$

check the domain, only 19° is in the domain

$$\therefore \theta = 49^\circ$$

(d) Given $f(x) = \begin{cases} 1-x^2, & \text{for } x \leq 1 \\ kx+1, & \text{for } x > 1 \end{cases}$

(i) Find the value of k for which $f(x)$ is continuous at $x = 1$.

1

$$f(1) = 1 - 1^2$$

$$= 0$$

for the function to be continuous at $x = 1$,

$$k(1) + 1 = 0$$

$$\therefore k = -1$$

(ii) If $k = -1$, find the value of $f(-3) + f(2)$.

1

$$\text{if } k = -1, f(x) = \begin{cases} 1-x^2, & \text{for } x \leq 1 \\ 1-x, & \text{for } x > 1 \end{cases}$$

$$f(-3) = 1 - (-3)^2$$

$$= -8$$

$$f(2) = 1 - (2)$$

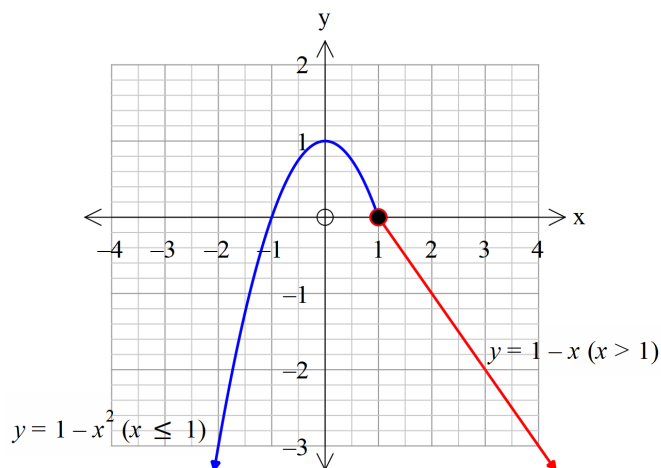
$$= -1$$

$$\therefore f(-3) + f(2) = -8 - 1$$

$$= -9$$

(iii) If $k = -1$, find the range of $f(x)$.

1



$\therefore$ the range of $f(x)$ is $y \leq 1$.

- (e) If the minimum value of parabola $y = x^2 - 3x + k$ is 5, find the value of k .
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$$y = \left(x - \frac{3}{2}\right)^2 - \frac{9}{4} + k$$

Or: axis of symmetry at $x = \frac{-b}{2a} = \frac{3}{2}$

substitute $x = \frac{3}{2}$ into $5 = x^2 - 3x + k$

$$5 = \left(\frac{3}{2}\right)^2 - 3\left(\frac{3}{2}\right) + k$$

The minimum value $-\frac{9}{4} + k = 5$

$$\therefore k = \frac{29}{4}$$

$$k = 5 + \frac{9}{4} = \frac{29}{4}$$

Question 13 (12 marks)

- (a) Differentiate the following functions, leaving your answers in simplest factorised form where possible:

(i) $y = \frac{2}{\sqrt{x}}$ 2

$$y = 2x^{-\frac{1}{2}}$$

$$y' = 2\left(-\frac{1}{2}\right)x^{-\frac{3}{2}}$$

$$y' = \frac{-1}{x\sqrt{x}}$$

(ii) $y = (e^{2x} - 1)^3$ 2

$$y' = 3(e^{2x} - 1)^2 \times 2e^{2x}$$

$$y' = 6e^{2x}(e^{2x} - 1)^2$$

(iii) $f(x) = x^3\sqrt{x^2 - 1}$ 2

$$f'(x) = 3x^2\sqrt{x^2 - 1} + \frac{1}{2}(x^2 - 1)^{-\frac{1}{2}}x^3$$

$$= 3x^2\sqrt{x^2 - 1} + \frac{x^3}{2\sqrt{x^2 - 1}}$$

$$= \frac{3x^4 + x^3 - 3x^2}{2\sqrt{x^2 - 1}}$$

(b) Consider the function $f(x) = \frac{1}{3}x^3 - x^2 - 15x + 3$.

(i) Find $f'(x)$.

1

$$f'(x) = x^2 - 2x - 15$$

(ii) Find the values of x such that $f'(x) = 0$.

2

$$x^2 - 2x - 15 = 0$$

$$(x - 5)(x + 3) = 0$$

$$\therefore x = 5, -3$$

(iii) Find the set of values of x for which $f(x)$ is decreasing.

1

$$(x - 5)(x + 3) < 0 \text{ for decreasing function}$$

$$\therefore -3 < x < 5$$

(iv) Find the gradient of the tangent and normal of $f(x)$ at $x = 0$.

2

$$x = 0, f'(0) = -15$$

$$\therefore m_{\text{tangent}} = -15$$

$$m_{\text{normal}} = \frac{1}{15}$$

Question 14 (11 marks)

(a) Consider the equation $x + \frac{2}{x} = p$, find the values of p for which the equation has real roots. 2

$$x^2 + 2 = px$$

$$x^2 - px + 2 = 0$$

$$\Delta \geq 0 \text{ for real roots}$$

$$(-p)^2 - 4(1)(2) \geq 0$$

$$p^2 - 8 \geq 0$$

$$p = \pm 2\sqrt{2}$$

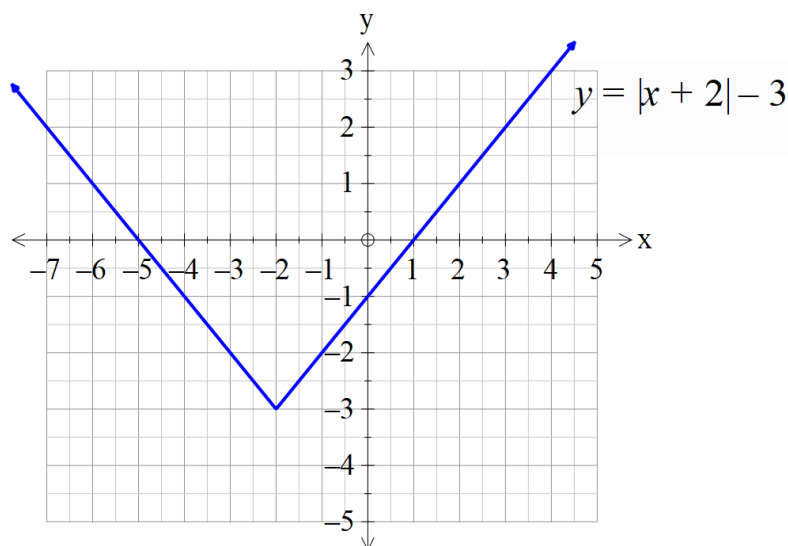
$$\therefore p \leq -2\sqrt{2}, p \geq 2\sqrt{2}$$

Comment: well done in the first section but poorly done at the end (last line incorrect).

(b) For $f(x) = |x + 2| - 3$,

(i) Sketch the function, showing any intercepts.

3



Comment: well done!

(ii) Hence, or otherwise, find the values for $f(x) \geq 1$.

1

Draw $y = 1$ on the above graph,

for $f(x) \geq 1$, $x \leq -6, x \geq 2$.

or solve $|x + 2| - 3 \geq 1$ algebraically:

$$|x + 2| \geq 4$$

$$x + 2 \leq -4, x + 2 \geq 4$$

$$\therefore x \leq -6, x \geq 2$$

Comment: mostly well done!

(c) A bacteria replicated itself according to the equation $N(t) = 140 \times 2^{3.5t}$, where $N(t)$ is the number of bacteria present in a test tube t hours after the experiment began.

(i) What is the initial number of bacteria added to the test tube?

1

$$N(0) = 140 \times 2^0$$

$$= 140$$

Comment: well done!

- (ii) How many bacteria are there after 10 hours?

2

Answer in scientific notation correct to 3 significant figures.

$$\begin{aligned}N(0) &= 140 \times 2^{3.5 \times 10} \\&= 140 \times 2^{35} \\&= 4.81 \times 10^{12} \text{ (3 sig. fig.)}\end{aligned}$$

Comment: well done!

- (iii) When the number of bacteria reaches 3×10^{25} in the test tube, it will stop replicating. How long before this number is reached?

Give your answer correct to the nearest minute.

2

$$\begin{aligned}3 \times 10^{25} &= 140 \times 2^{3.5t} \\ \frac{3 \times 10^{25}}{140} &= 2^{3.5t} \\ \ln\left(\frac{3 \times 10^{25}}{140}\right) &= \ln(2^{3.5t}) \\ 3.5t &= \frac{\ln\left(\frac{3 \times 10^{25}}{140}\right)}{\ln 2} \\ t &= \frac{\ln\left(\frac{3 \times 10^{25}}{140}\right)}{3.5 \ln 2} \\ &= 22.1439... \text{ hours} \\ &= 22 \text{ hours } 9 \text{ minutes (nearest minutes)}\end{aligned}$$

Comment: generally well done, however, some students used hours instead of minutes.

Question 15 (11 marks)

- (a) Given that $f(x) = \frac{2x+1}{3x-4}$, $\left(x \neq \frac{4}{3}\right)$, find $f'(-2)$.

2

$$f'(x) = \frac{(3x-4)(2) - 3(2x+1)}{(3x-4)^2}$$

$$= \frac{6x-8-6x-3}{(3x-4)^2}$$

$$= \frac{-11}{(3x-4)^2}$$

$$f'(-2) = \frac{-11}{(3(-2)-4)^2}$$

$$= \frac{-11}{100}$$

$$\text{or: } f(x) = (2x+1)(3x-4)^{-1}$$

$$f'(x) = (3x-4)^{-1} \times 2 + (2x+1) \times (-1)(3x-4)^{-2} \times 3$$

$$= \frac{2}{3x-4} - \frac{3(2x+1)}{(3x-4)^2}$$

$$= \frac{2(3x-4) - 6x - 3}{(3x-4)^2}$$

$$= \frac{-11}{(3x-4)^2}$$

Comments: lots of arithmetic errors. 1 mark for correct differentiation, 1 mark for correct answer.

- (b) Solve $\log_2(x+4) + \log_2(x-1) - \log_2(x+2) = 2$.

3

$$\log_2 \left(\frac{(x+4)(x-1)}{(x+2)} \right) = 2$$

$$4 = \frac{(x+4)(x-1)}{(x+2)}$$

$$4x+8 = x^2 + 3x - 4$$

$$x^2 - x - 12 = 0$$

$$(x-4)(x+3) = 0$$

$$x = 4, -3$$

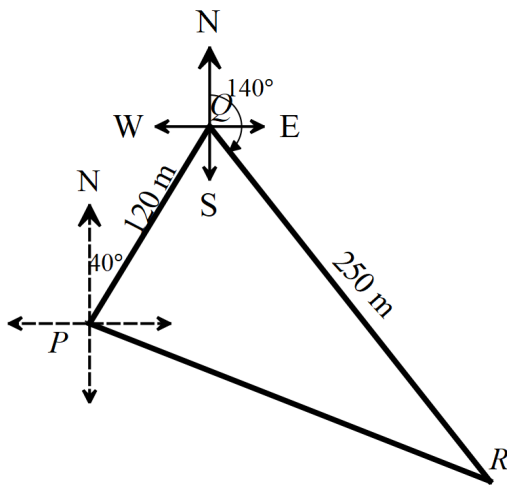
Check the solutions: substitute $x = -3$ into the original equation,

$\log_2(-3+4) + \log_2(-3-1) - \log_2(-3+2)$ has no solution as $\log_2(-3-1)$ is undefined.

$\therefore x = 4$ is the only solution.

Comments: 1 mark for using log laws to find the quadratic, 1 mark for solving quadratic, 1 mark for correct answer. Many students missed the third mark.

- (c) A farmer walked 120 metres from P to Q on a bearing of 040°T .
He then walked 250 metres from Q to R on a bearing of 140°T .



- (i) Find the value of $\angle PQR$.

1

$$\angle PQR = 80^\circ \text{ (use angles on parallel lines to deduce)}$$

- (ii) Find the length of PR , correct to the nearest metre.

2

$$\begin{aligned} PR^2 &= 120^2 + 250^2 - 2 \times 120 \times 250 \cos 80^\circ \\ PR &= \sqrt{120^2 + 250^2 - 2 \times 120 \times 250 \cos 80^\circ} \quad (PR > 0) \\ &= 257.839... \\ &= 258 \text{ m (nearest metre)} \end{aligned}$$

Comments: some students were working in radians and got an answer of 289.0038661.

- (iii) Find the bearing of R from P , correct to the nearest degree.

2

$$\begin{aligned} \frac{\sin \angle QPR}{250} &= \frac{\sin 80^\circ}{258} \\ \sin \angle QPR &= \frac{250 \sin \angle QPR}{258} \\ \angle QPR &= \sin^{-1} \left(\frac{250 \sin \angle QPR}{258} \right) \\ &= 72^\circ 39' \\ \therefore \text{ the bearing of } R \text{ from } P &= 72^\circ 39' + 40^\circ = 112^\circ 39'. \end{aligned}$$

Comments: some students found the angle but failed to find the bearing.

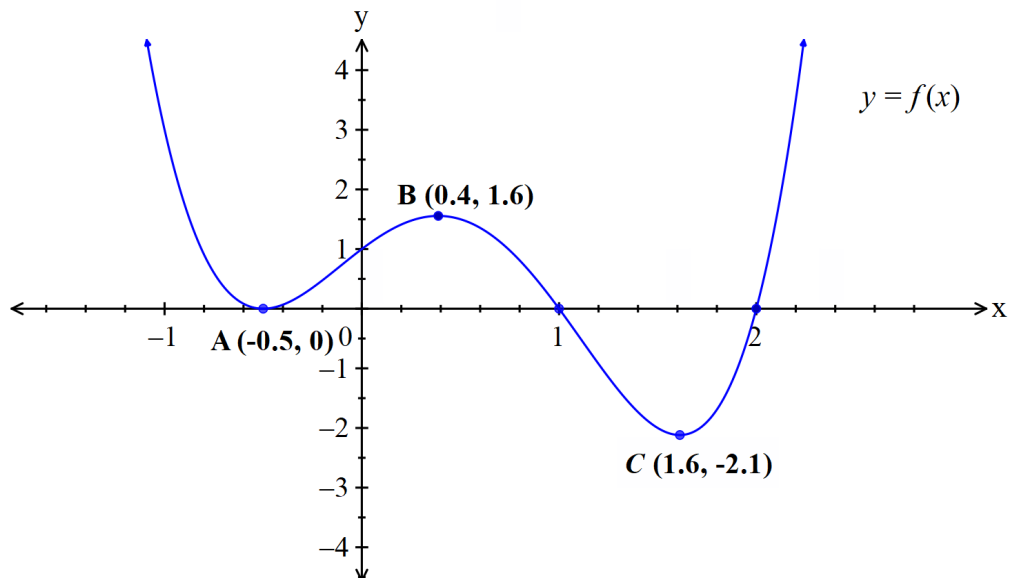
(iv) Find the area of $\triangle PQR$. Answer correct to the nearest m^2 .

1

$$\begin{aligned} \text{Area} &= \frac{1}{2}(120)(250)\sin 80^\circ \\ &= 14772 \text{ m}^2 \text{ (nearest m}^2\text{)} \end{aligned}$$

Question 16 (11 marks)

- (a) The diagram below shows the graph of a function $y = f(x)$. Point A, B and C are stationary points at which the derivatives are zero.

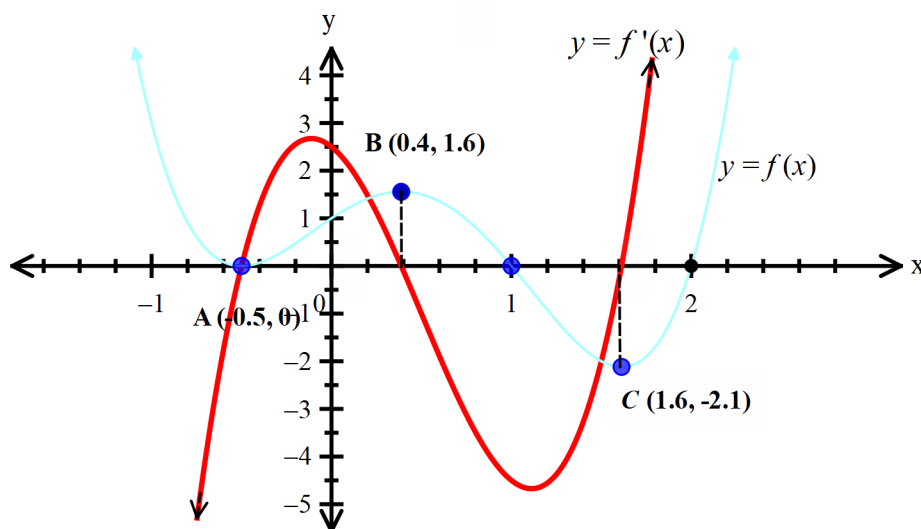


Comments:

- Many students need to learn to draw their derivative graphs correctly knowing what each key feature represent in the derivative function.
- Label the x and y axis correctly on the derivative graph.
- Scales need to be evenly distributed.

- (i) Sketch the graph of $f'(x)$. Clearly label all essential information.

3



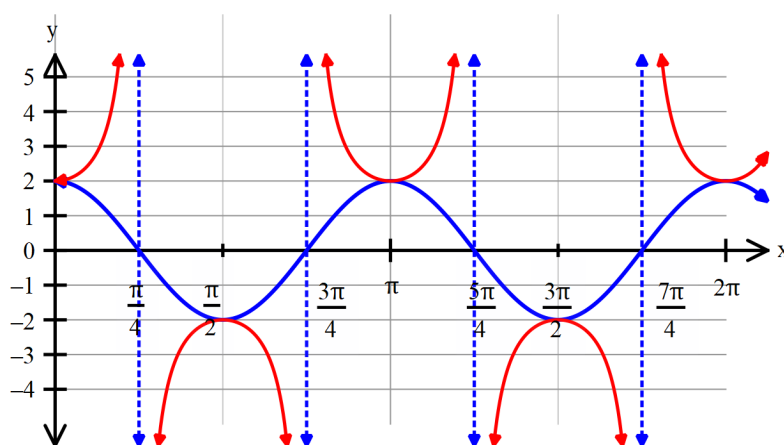
- (ii) State the values of x for which the derivative of $f(x)$ is increasing.

1

$$-0.5 < x < 0.4, \quad x > 1.6$$

This question has been excluded from this assessment as there is a mistake in the question. The intended question should be : State the values of x for which $f(x)$ is increasing. The total mark and student marks have been adjusted.

- (b) The graph of $y = 2 \cos(2x)$ is shown below, where x is in radians.



Comment:

- Many students do not understand the concept of amplitude and period.
- Many mistaken $y = 2 \sec 2x$ being the reciprocal graph of $y = 2 \cos 2x$
- Many need to learn to solve their equations in radians and be comfortable with it.

- (i) Find the period and amplitude. 2

The amplitude is 2.

The period is π .

- (ii) On the above graph, label all the x - and y - intercepts on both axes. 2

- (iii) Draw $y = 2\sec(2x)$ on the above graph, clearly showing any asymptotes. 1

- (iv) Find the exact solutions for $2\sec(2x) = 4$ for $0 \leq x \leq 2\pi$. 2

$$\sec(2x) = 2$$

$$\cos(2x) = \frac{1}{2}$$

$$0 \leq x \leq 2\pi$$

$$0 \leq 2x \leq 4\pi$$

$$2x = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}$$

$$x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

Comment: Many did not adjust the domain to accommodate the multiple angle, have missed the other 2 solutions.

End of Examination

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